

Predicting Human-Perceived Image Quality from Real-World Photos

Team member: Muhammad Ammar Sohail (202356790)
Second teammate: to be confirmed (GitHub: mkamaleldin7)

Project summary

This project asks whether a small transfer-learning model can estimate how technically good a photograph looks to people. The model receives one photograph and predicts a single quality score. It does not need a clean reference image to compare against, so this is called blind or no-reference image quality assessment.

Task type	Input	Output
Regression	One RGB photograph	One continuous MOS quality score

Why this problem matters

Photo upload and media systems often receive images that are out of focus, noisy, poorly exposed, or heavily compressed. A quality estimate could help prioritize images for review, recommend a retake, or rank uploads when no original reference image is available. The project is useful without claiming that a course model would replace human judgment.

Planned system

Step	Simple plan
1. Read the image	Use the released 512x384 RGB image.
2. Predict quality	Use an ImageNet-pretrained ResNet18 with a one-value regression head.
3. Compare with people	Select the best checkpoint using SROCC on the validation set.

The goal is a clean applied deep-learning experiment, not a new image-quality method. The milestone work below is limited to inspecting the data and fixing the evaluation plan before training.

2. DATASET INSPECTION

KoniQ-10k dataset

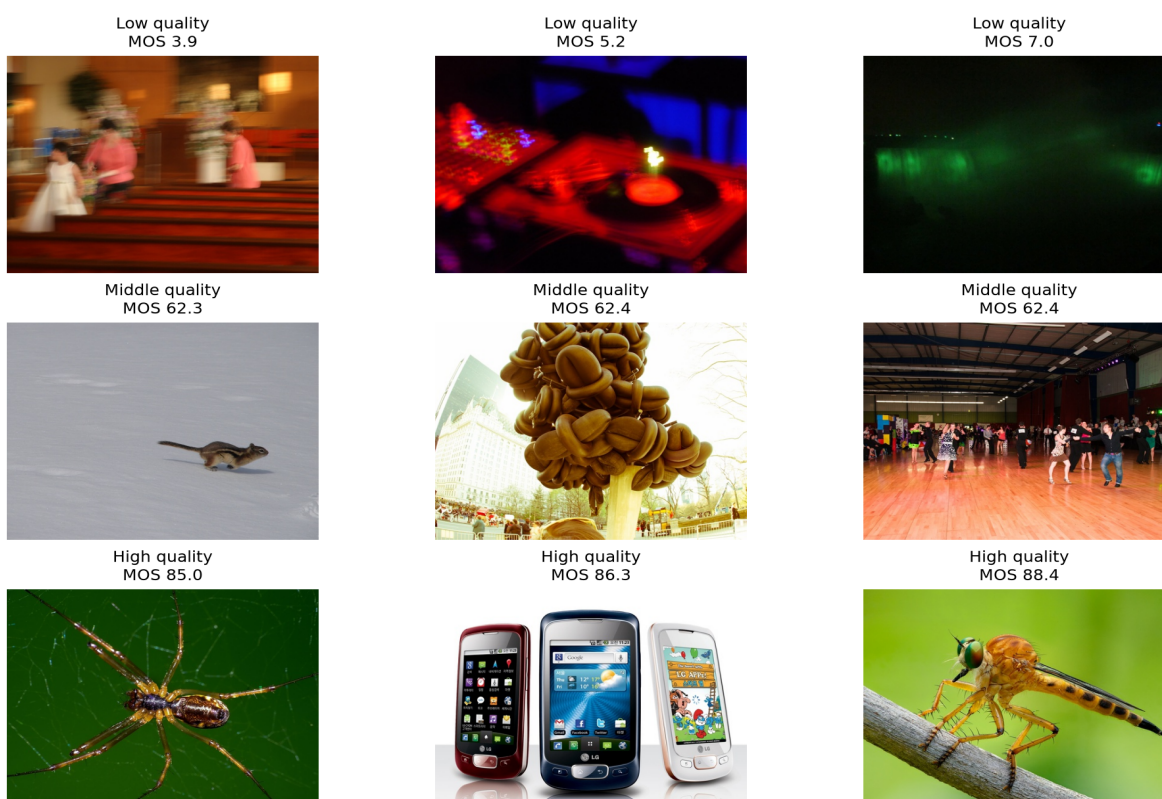
KoniQ-10k contains 10,073 real-world photographs selected from Flickr-related public image data. The images were rated by people for technical quality. Unlike a dataset made by adding one artificial blur to each clean image, these photographs contain natural combinations of issues such as blur, noise, exposure problems, and compression artifacts.

Dataset fact	Observed value
Images	10,073 metadata-matched images (300 extra archive files ignored)
Released image size	512x384 RGB version used for this plan
Target	Released MOS score on a 0-100 scale
Target range	3.91 to 88.39; mean 58.73; median 62.35
Metadata checks	No missing metadata values or duplicate filenames

What the target means

People gave ratings on a 1-5 scale. The released metadata stores the derived mean opinion score (MOS) after rescaling it to 0-100. We use the released MOS as a continuous target: a lower value means lower perceived technical quality. This is why the task is regression rather than categories such as low, medium, and high.

Representative KoniQ-10k images across the released MOS range



MOS is the released 0-100 target; lower values indicate lower perceived technical quality.

Figure 1. Actual KoniQ-10k images selected from the low, middle, and high parts of the MOS range.

Access and licensing: the 512x384 archive used for the audit is listed as CC BY 4.0 on Zenodo. The underlying source photographs can have individual attribution requirements, so the repository contains code and figures only and does not redistribute the full image archive.

3. EXPERIMENTAL PLAN

Split, metric, and planned comparison

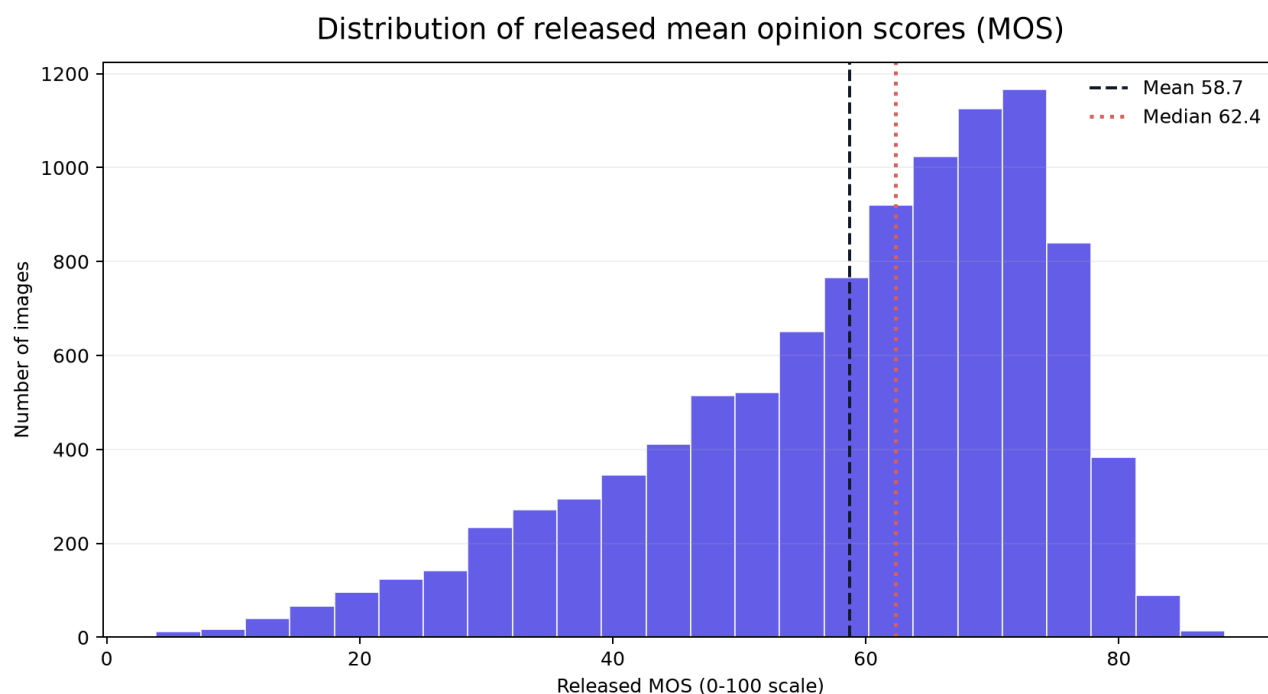


Figure 2. Distribution of the released MOS target across all 10,073 images.

Train, validation, and test split

Split	Images	Use
Training	7,058	Fit the model
Validation	1,000	Choose the checkpoint and compare planned variants
Test	2,015	Use once for the final report

Single validation metric: Spearman rank correlation (SROCC)

SROCC measures whether the model orders images in a similar way to human ratings. This fits the goal better than accuracy because the target is continuous and the difference between two images' exact scores is less important than whether the better-looking image is ranked higher. The test set will remain untouched until the end.

Small set of later experiments

Comparison	Question
Mean-score baseline vs. ResNet18	Does the image model learn more than always predicting the training mean?
Frozen backbone vs. limited fine-tuning	Does adapting the final visual features help?
Optional: 224x224 vs. 512x384	Does preserving more image detail help quality prediction?

Limitations

Human quality judgments are subjective, and the dataset comes from a particular online-photo collection. The project will therefore make claims about performance on KonIQ-10k rather than about every photograph or every viewer. The proposal also treats technical quality as the target, not artistic beauty.

Sources: KonIQ-10k paper (Hosu et al., 2020), authors' metadata repository, and the 512x384 Zenodo data record. Supporting code and exact run instructions are included in the repository README.